

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
IV B. TECH I SEMESTER

BATCH 2023-27	MAJOR PROJECT 2026-27	ACADEMIC YEAR 2026-27
--------------------------------	--	--

Batch Id : 23MPCS-B17
Title of the Project : ResumeXAI: An Explainable LLM-RAG Framework for Semantic Resume Analysis, Information Extraction, and Job Matching

TEAM MEMBERS

S. No.	Roll Number	Student Name	Mail Id	Contact Number
1	23881A05B4	RANGU SIRI SAHITHYA	sirisahithyamahesh@gmail.com	7207663880
2	24885A0509	MYADAM VINAYAK	vinayakvmeg@gmail.com	6304773815
3	24885A0511	GADDAM YASHWANTH	gaddamyashwanth1508@gmail.com	8019955410

GUIDE DETAILS

Name of the Guide	:	Mr.S.SURENDHAR
Designation	:	Assistant Professor
Department	:	CSE
Mail Id	:	surendhar1673@vardhaman.org
Contact Number	:	8610895394
Area of interest	:	Deep Learning-NLP

*Signature of the
Projects In charge*

*Signature of the
Guide with Date*

*Signature of the
Project Coordinator*

Batch Id : 23MPCS-B17

Title of the Project : ResumeXAI: An Explainable LLM-RAG Framework for Semantic Resume Analysis, Information Extraction, and Job Matching

ABSTRACT

AI Resume Analyzer is an intelligent web-based platform designed to automate and enhance the process of resume evaluation for both job seekers and recruiters. The platform leverages modern artificial intelligence technologies, particularly Large Language Models (LLMs), to parse, analyze, and score resumes against job descriptions in real time. Built using a scalable full-stack architecture, the system integrates powerful AI components including natural language processing (NLP), semantic similarity matching, structured information extraction, Retrieval-Augmented Generation (RAG), and Explainable Artificial Intelligence (XAI) into a unified resume analysis pipeline. This architecture enables the system to accurately interpret resume content, identify skill gaps, match candidate profiles to job requirements, generate ATS compatibility scores, and provide context-aware recommendations for resume improvement. A key enhancement of the proposed system is the Explainable AI Visualization module, which improves transparency by presenting section-wise score breakdowns, feature contribution analysis, interactive visual dashboards, resume heatmaps, and improvement impact predictions, enabling users to clearly understand the factors influencing their ATS score and prioritize resume enhancements effectively. The system evaluates resumes across multiple dimensions, including skills match, experience relevance, educational qualifications, project quality, keyword optimization, and resume formatting, while delivering near-instant analysis results through a responsive web interface that supports PDF and DOCX uploads. It also allows recruiters to compare resumes against job descriptions for efficient candidate screening. By combining AI-driven analysis with transparent and interpretable visual explanations, the proposed platform enhances user trust, supports informed decision-making, and provides an open, extensible, scalable, and production-ready solution for intelligent resume evaluation, career guidance, and modern recruitment applications.

Keywords: Resume Analysis, Large Language Models, Natural Language Processing, Semantic Matching, Skill Extraction, Job Description Matching, RAG, FastAPI, React, pgvector, Microservices.

INTRODUCTION

OVERVIEW

The AI Resume Analyzer with Explainable AI Visualization is an intelligent web-based platform developed to automate and enhance the resume evaluation process for job seekers and recruiters using Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs). The system allows users to upload resumes in PDF or DOCX format and compare them with job descriptions to assess Applicant Tracking System (ATS) compatibility by analyzing skills, experience, education, projects, resume formatting, and keyword relevance. It employs semantic similarity matching, structured information extraction, and Retrieval-Augmented Generation (RAG) to identify skill gaps, generate personalized improvement suggestions, and provide context-aware career guidance. As a key enhancement over existing resume analyzers, the proposed system incorporates an Explainable AI (XAI) Visualization module that presents transparent ATS score explanations through section-wise score breakdowns, feature contribution analysis, resume heatmaps, interactive dashboards, and improvement impact predictions, enabling users to understand the reasoning behind their ATS score and effectively improve their resumes. Built on a scalable full-stack architecture with a responsive web interface, the platform delivers accurate, efficient, and interpretable resume analysis, making it a reliable solution for intelligent career guidance and modern recruitment applications.

RELEVANCE OF PROJECT

With the increasing adoption of AI-based Applicant Tracking Systems (ATS) in recruitment, optimizing resumes has become essential for job seekers. Existing resume analyzers often provide only an ATS score without explaining the factors influencing it, limiting users' ability to improve their resumes effectively. The proposed AI Resume Analyzer with Explainable AI Visualization addresses this gap by providing transparent score breakdowns, visual insights, and personalized recommendations. This helps users understand their resume's strengths and weaknesses, improve ATS compatibility, and make informed enhancements. The system also supports recruiters by enabling faster, more accurate, and transparent candidate evaluation.

LITERATURE REVIEW

S.NO	YEAR	TITLE OF PAGE	JOURNAL/ CONFERENCE	CONTRIBUTION	RESEACH GAP
1	2026	AI Resume Analyzer: A Survey and Implementation Report on Intelligent Resume Evaluation Using Large Language Models	International Journal of Pharmacy and Medicine Sciences (IJPMS)	Full-stack AI resume analyzer using LLMs, semantic matching, ATS scoring, suggestions.	Limited personalization no recruiter simulation, lacks explainability
2	2026	Resume Analyzer and ATS Scoring System	Journal of Academic Trends and Innovative Research (JATIR)	Student-oriented ATS scoring platform with resume parsing and optimization	Mostly keyword-based, limited semantic reasoning and recruiter insights.
3	2026	AI Resume Builder and Analyzer	International Journal of Multidisciplinary Research in Science, Engineering and Technology	Combines resume builder, ATS optimization, Gemini/Ollama integration.	No adaptive learning, weak analytics, limited multilingual support
4	2026	ResuScan: AI-Based Resume Parser with ATS Score	IJPREMS	Resume parser, job description comparison, ATS compatibility score	No interview readiness, no portfolio or GitHub evaluation
5	2025	AI Hiring with LLMs: A Context-Aware and Explainable Multi-Agent Framework for Resume	CVPR	Multi-agent architecture with RAG and explainable evaluation.	Research prototype; lacks end-user career coaching and deployment focus

		Screening			
6	2025	Layout-Aware Parsing Meets Efficient LLMs	arXiv	Robust parsing of complex resume layouts using layout-aware models.	Focuses on extraction, not user-facing ATS improvement features
7	2025	Beyond Human Labels: Resume Bench	Aclanthology	Benchmark for evaluating LLMs on multilingual resume parsing	Benchmark only, not a complete ATS assistant or recommendation system.
8	2025	FAIRE: Assessing Racial and Gender Bias in AI-Driven Resume Evaluations	arXiv	Measures fairness and bias in AI hiring models	Highlights the need for bias-aware ATS tools

PROBLEM STATEMENT

Many qualified candidates are rejected by Applicant Tracking Systems (ATS) due to poorly optimized resumes, missing keywords, or improper formatting. Existing resume analyzers provide only ATS scores with limited explanations, making it difficult for users to improve their resumes. Therefore, an AI-powered and explainable resume analysis system is needed to provide accurate evaluations, transparent score explanations, and personalized improvement suggestions..

Objectives

1. Develop an AI-powered resume analyzer using NLP and LLMs.
2. Analyze resume sections and compare them with job descriptions.
3. Calculate ATS compatibility and identify skill gaps.

4. Generate personalized resume improvement suggestions.
5. Provide Explainable AI (XAI) visualizations for transparent ATS scoring.

Outcomes

1. Accurate and automated ATS-based resume evaluation.
2. Improved resume quality through AI-driven recommendations.
3. Transparent score explanations using Explainable AI.
4. Better job-resume matching by identifying missing skills..
5. Enhanced support for job seekers and recruiters in the hiring process..

REFERENCES:

[1] Mr. Ritesh Baral, & Prof. Saudamini Samantaray (2026). AI Resume Analyzer: A Survey and Implementation Report on Intelligent Resume Evaluation Using Large Language Models. https://www.researchgate.net/publication/406292358_AI_Resume_Analyzer_A_Survey_and_Implementation_Report_on_Intelligent_Resume_Evaluation_Using_Large_Language_Models

[2] Vandana swami, Nitin punia, , Gourav Kasana, Yogesh (2026)Resume Analyzer and ATS Scoring System. Journal of Academic Trends and Innovative Research (JATIR). Retrieved from https://jatir.org/publishedpapers/140579_PAPER.pdf

[3] Rutika Basatwar, Tejas Chaudhari, Prof. Bhushan Mali. (2026). AI Resume Builder and Analyzer. Retrieved from https://www.ijmrset.com/upload/24_AI%20Resume%20Builder%20and%20Analyzer.pdf

[4] Pratik Yawale, Shweta Bendke, & Prajakta Morey (2026). ResuScan: AI-Based Resume Parser with ATS Score. Retrieved from <https://www.ijprems.com/research-paper/resuscan-ai-based-resume-parser-with-ats-score>

- [5] Frank P.-W. Lo, Jianing Qiu, Zeyu Wang, Haibao Yu, Yeming Chen (2025) AI Hiring with LLMs: A Context-Aware and Explainable Multi-Agent Framework for Resume Screening. Retrieved from <https://arxiv.org/abs/2504.02870>
- [6] Fanwei Zhu, Jinke Yu, Zulong Chen, Ying Zhou, Junhao Ji, Zhibo Yang (2025). Layout-Aware Parsing Meets Efficient LLMs: A Unified, Scalable Framework for Resume Information Extraction and Evaluation. Retrieved from <https://arxiv.org/abs/2510.09722>
- [7] Zijian Ling, Han Zhang, Jiahao Cui, Zhequn Wu (2025). Beyond Human Labels: A Multi-Linguistic Auto-Generated Benchmark for Evaluating Large Language Models on Resume Parsing. Retrieved from <https://aclanthology.org/2025.emnlp-main.1626/>
- [8] Athena Wen, Tanush Patil, Ansh Saxena, Yicheng Fu, Sean O'Brien, Kevin Zhu (2025). FAIRE: Assessing Racial and Gender Bias in AI-Driven Resume Evaluations. Retrieved from <https://arxiv.org/abs/2504.01420>

Date of Submission:

*Signature of the
Guide with Date*

*Signature of the
HOD, CSE*